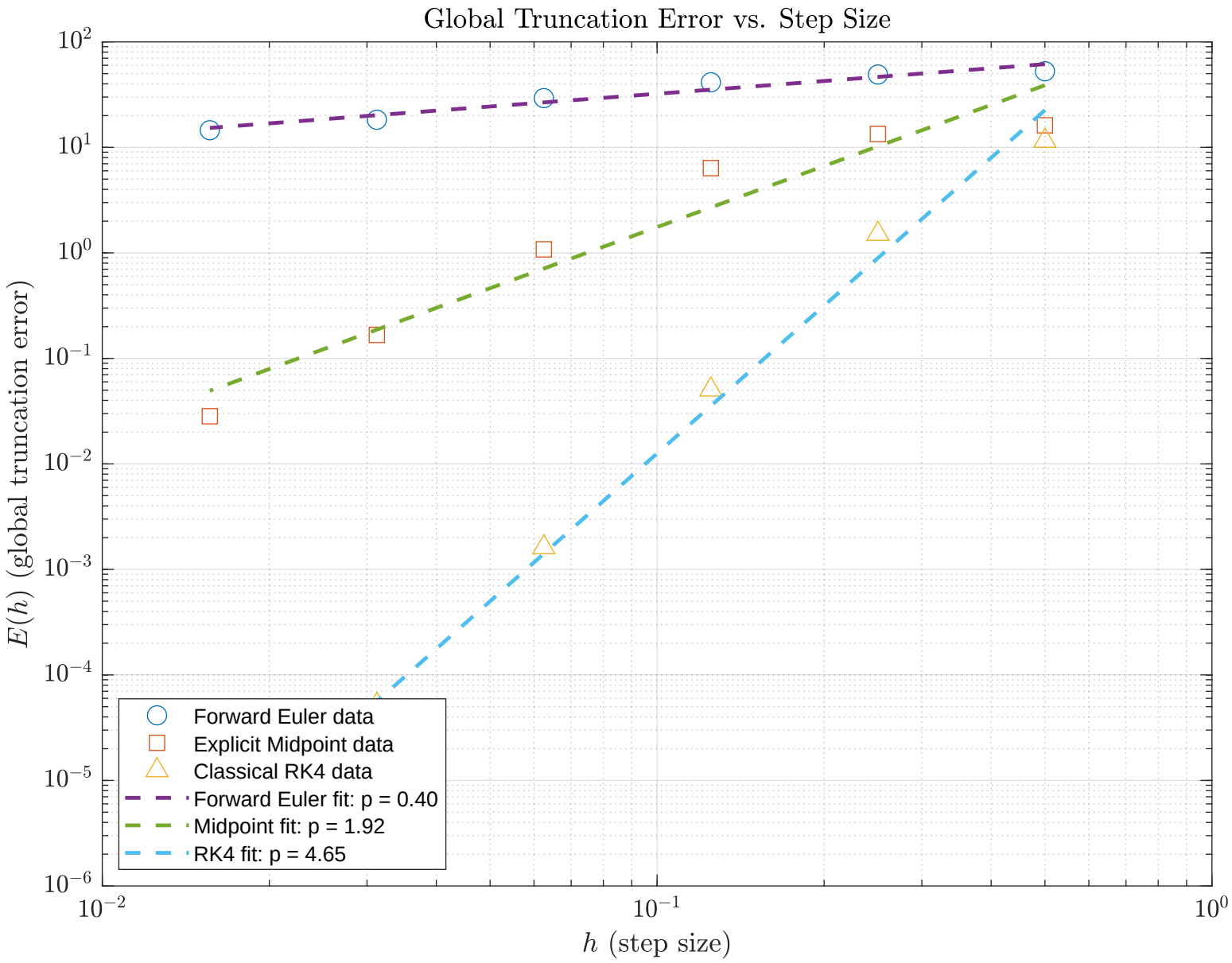

Global Truncation Error vs. Step Size
$E(h)$ (global truncation error)
h (step size)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-2}
10^{-1}
10^{0}
Forward Euler data
Explicit Midpoint data
Classical RK4 data
Forward Euler fit: p = 0.40
Midpoint fit: p = 1.92
RK4 fit: p = 4.65